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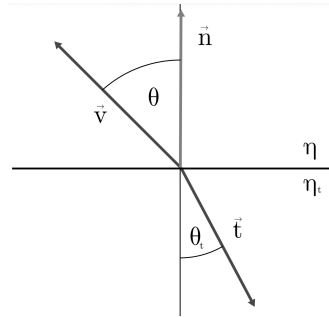
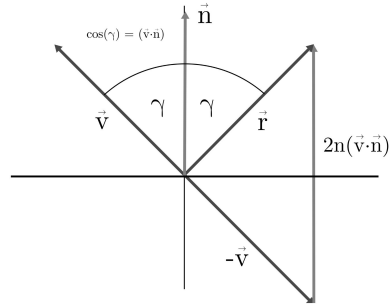
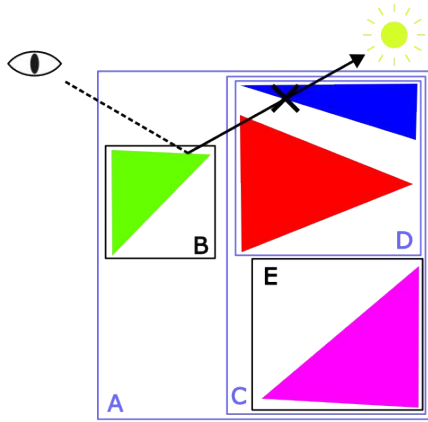
GPU Implementation of Acceleration Data Structures for Ray Tracing

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Supervisor: doc. Ing. Ivan Šimeček, Ph.D.

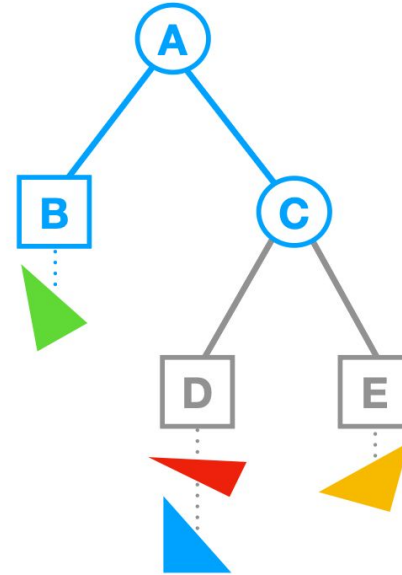
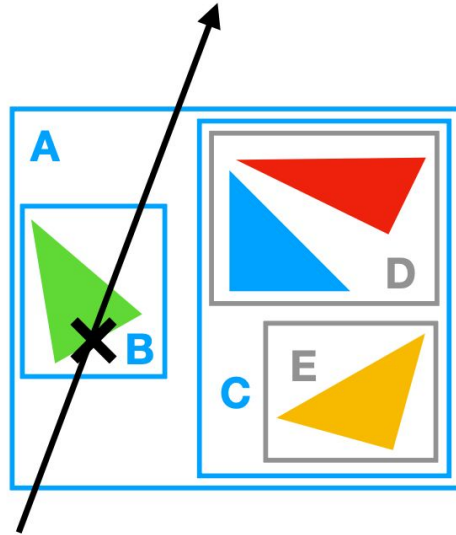
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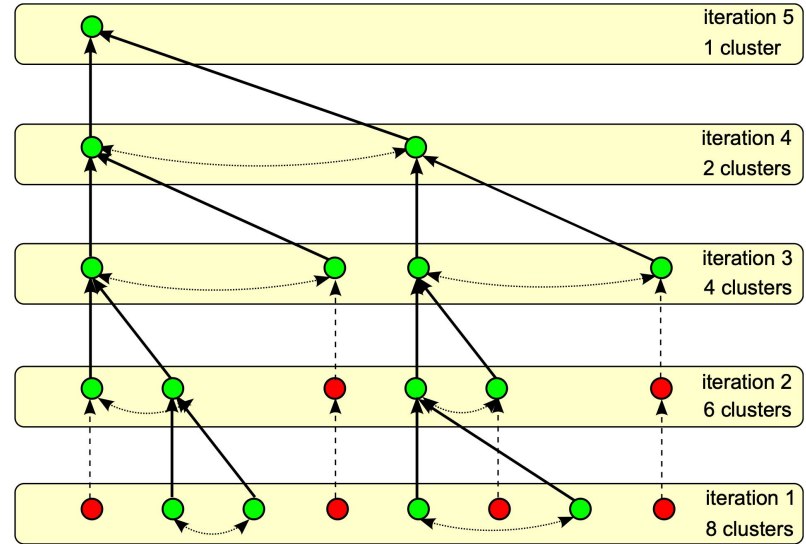
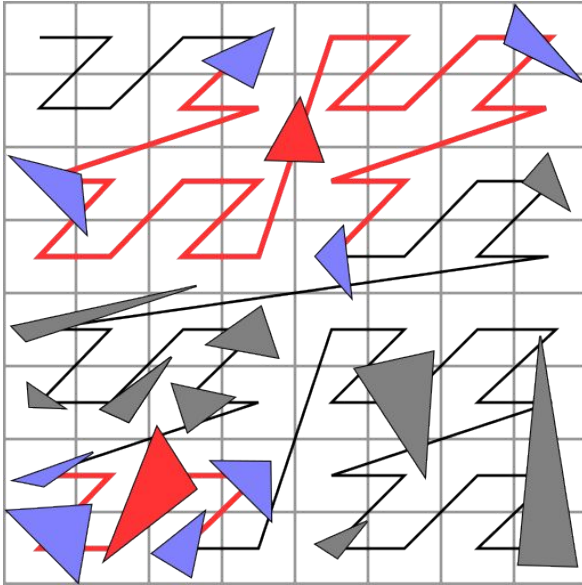
Ray tracing - Motivation



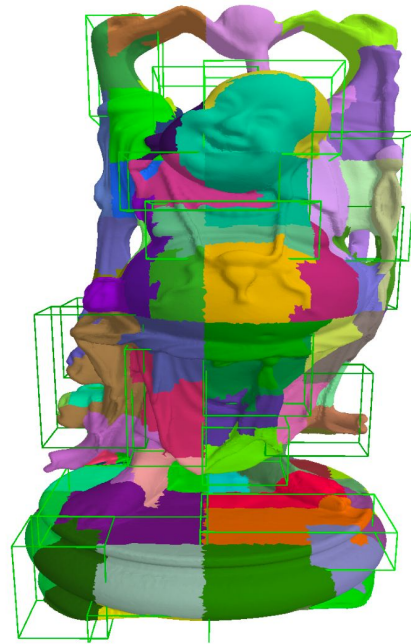
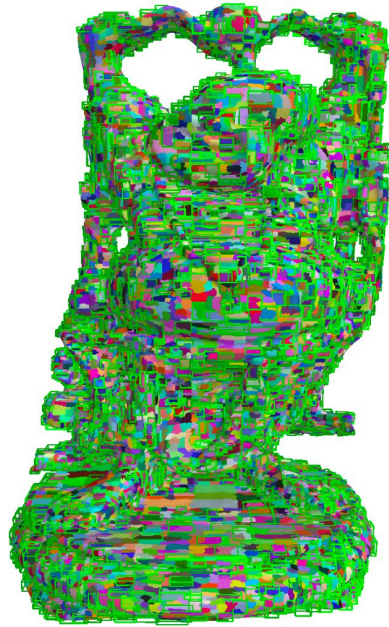
Bounding Volume Hierarchies (BVH)



BVH - Parallel Locally Ordered Clustering



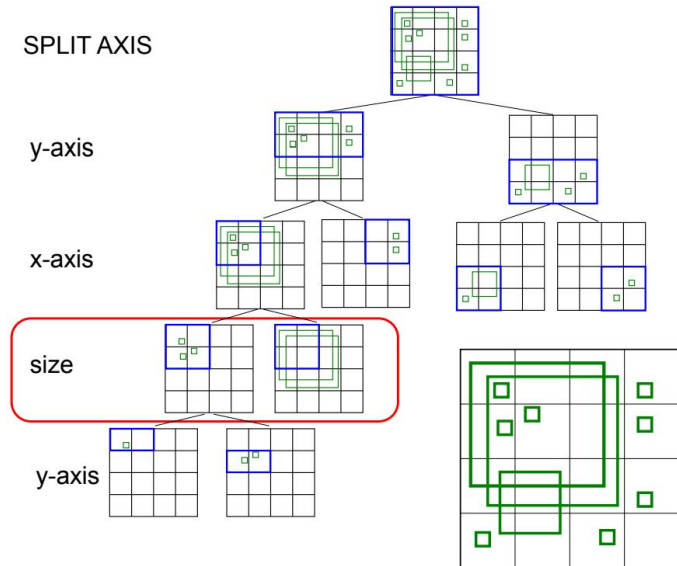
BVH - Parallel Locally Ordered Clustering



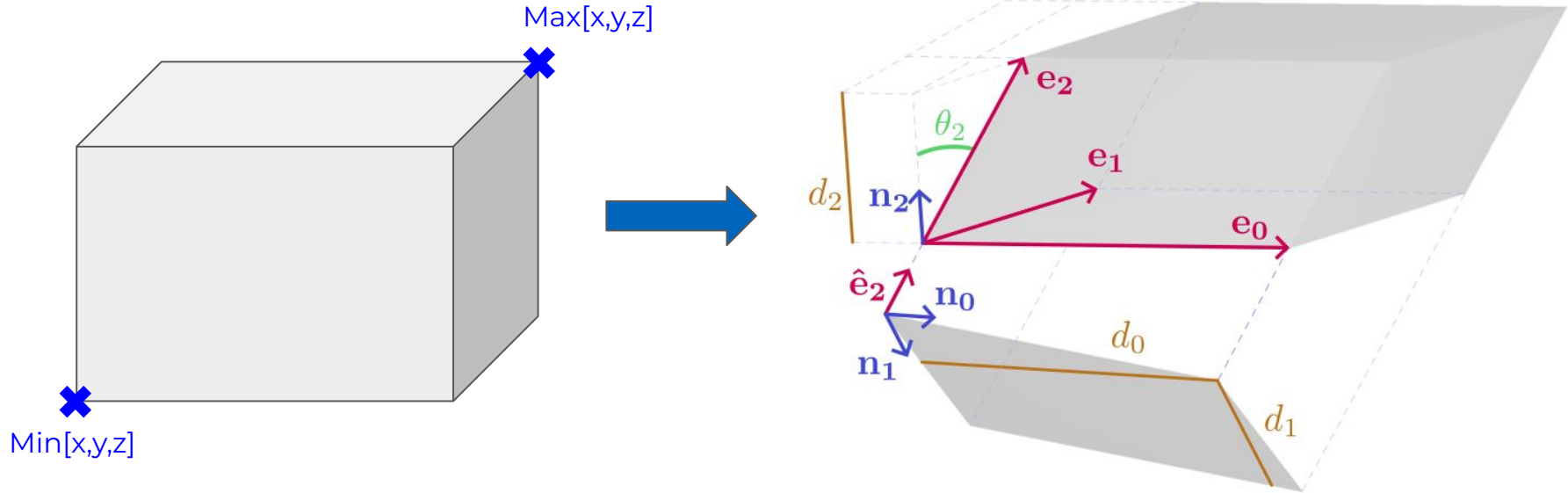
BVH - Extended Morton Codes (EMC)

xxyszyxxyszx

	00	01	10	11
00	0000	0001	0100	0101
01	0010	0011	0110	0111
10	1000	1001	1100	1101
11	1010	1011	1110	1111



BVH - Skewed Oriented Bounding Boxes (SOBB)



Goals

- PLOC (baseline)
- PLOC + EMC
- PLOC + SOBB
- PLOC + EMC + SOBB
- Comparison in a unified framework

Measurement Methodology

- Build time
- Trace performance [Mrays/s]
- BVH cost (Surface Area Heuristic)
- Trace time per unit of BVH cost

Implementation

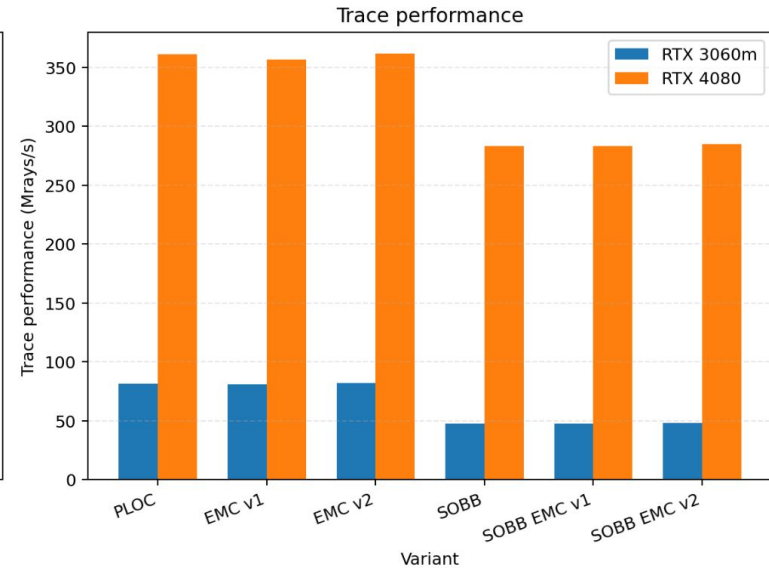
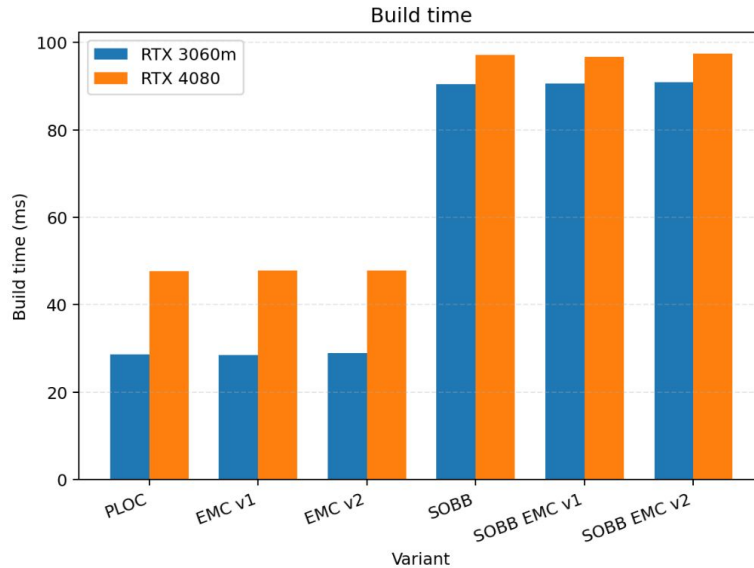


Implementation - Variants

- PLOC (nn search radius 25) - baseline
- PLOC EMC v1 (size bit, axes reordering)
- PLOC EMC v2 (size bit, axes reordering, variable bit count)
- PLOC SOBB (32-DOP, k^2 method)
- PLOC SOBB EMC v1
- PLOC SOBB EMC v2

Hardware

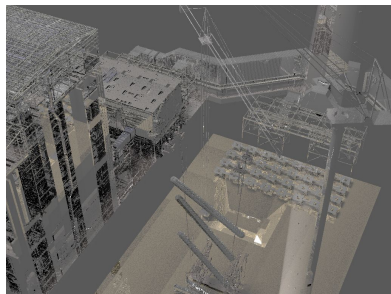
Nvidia GeForce RTX 3060m & Nvidia GeForce RTX 4080



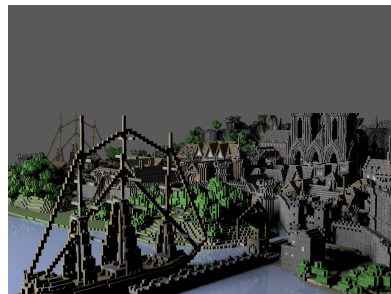
Experiments - Scenes



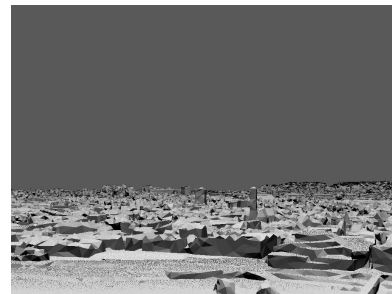
126k



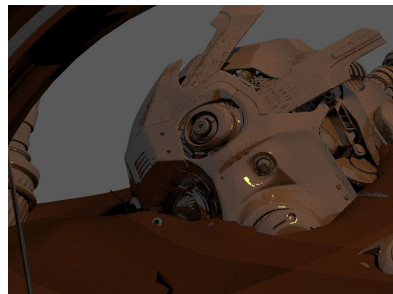
12.8M



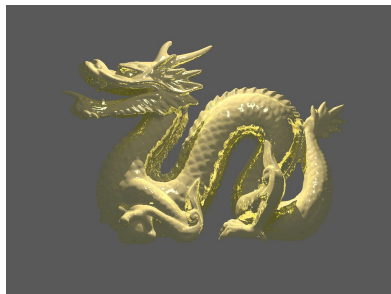
5.8M



1.3M



3.7M



800k



2.9M

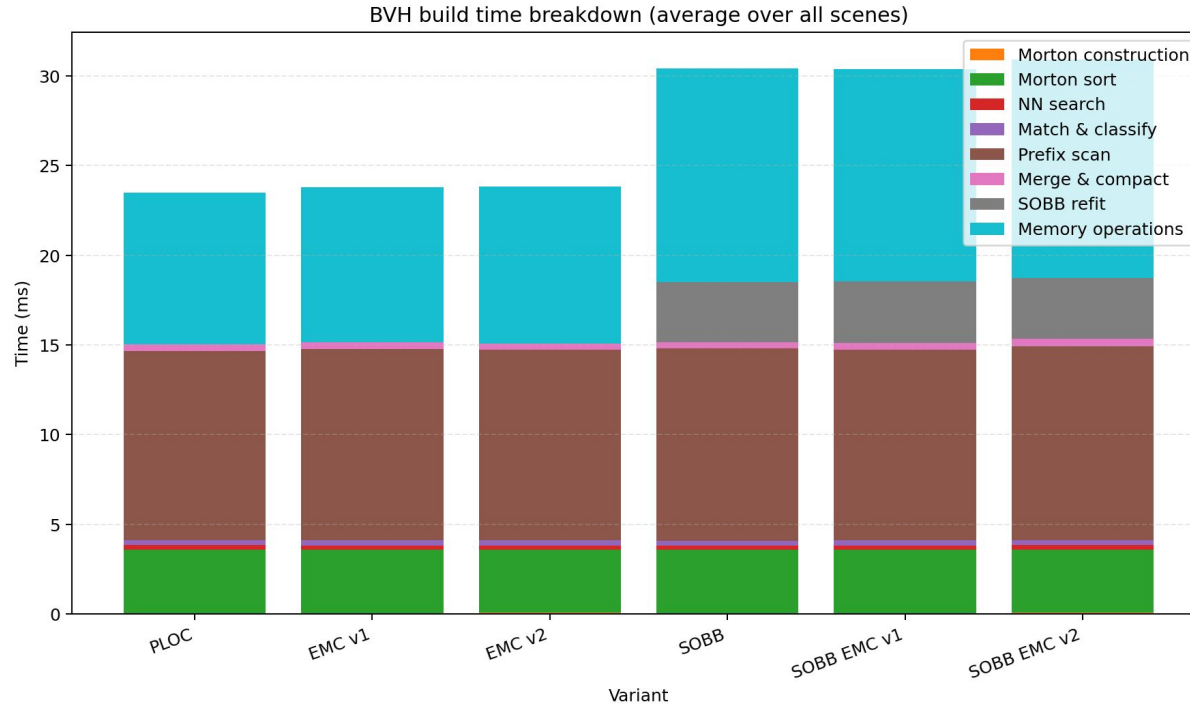


5.7M

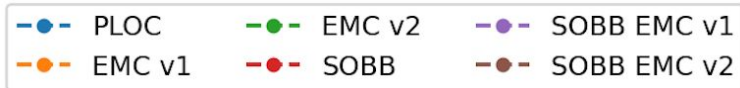
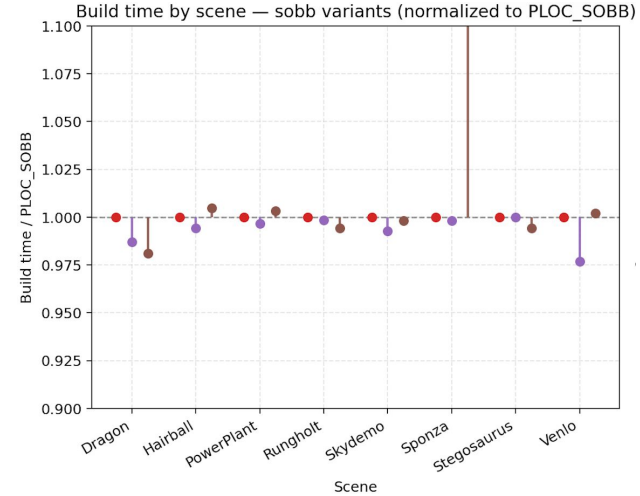
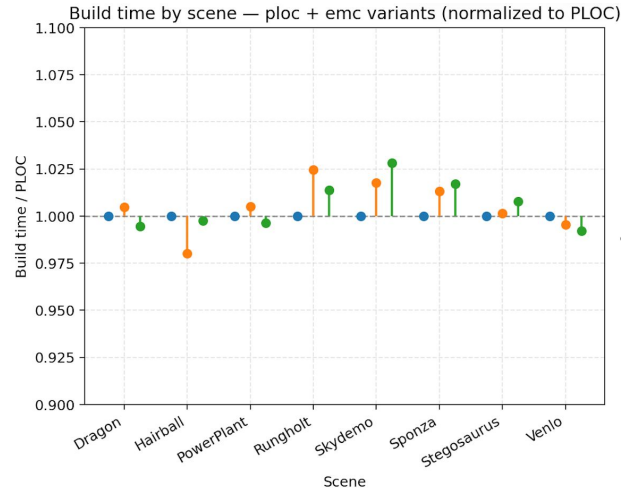
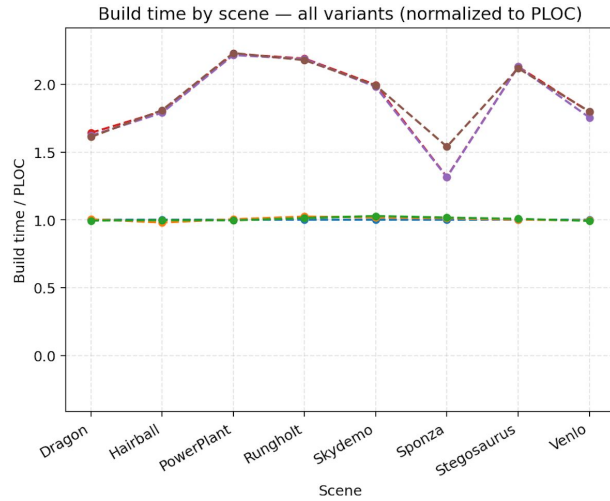
Experiments - Configuration

- Whitted style ray tracing
- Reflection and refraction rays based on Fresnel equations
- Maximum depth - 8
- Samples per pixel - 32

Results - Build time - Breakdown

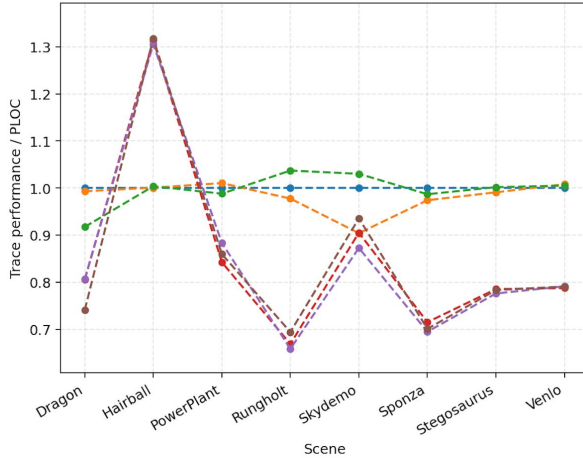


Results - Build time

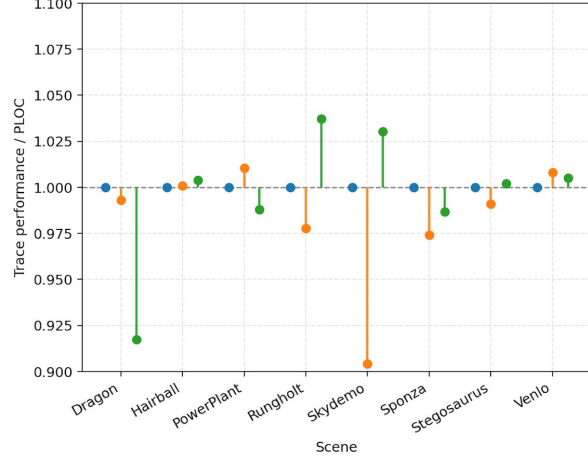


Results - Trace Performance

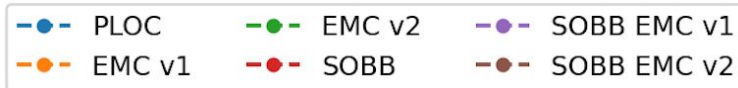
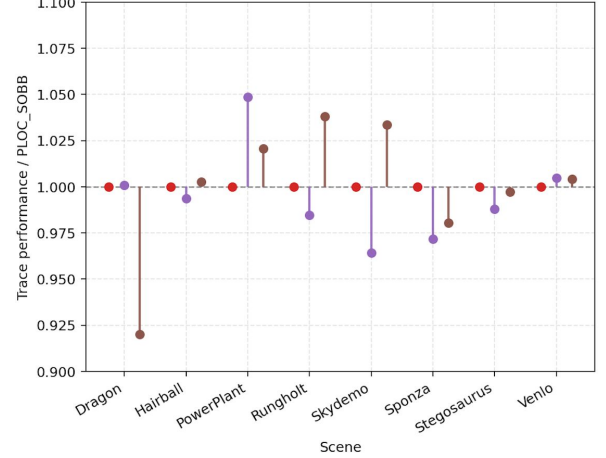
Trace performance by scene — all variants (normalized to PLOC)



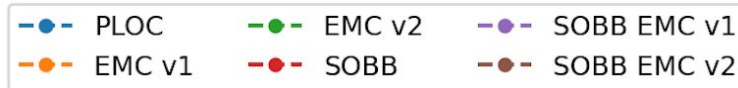
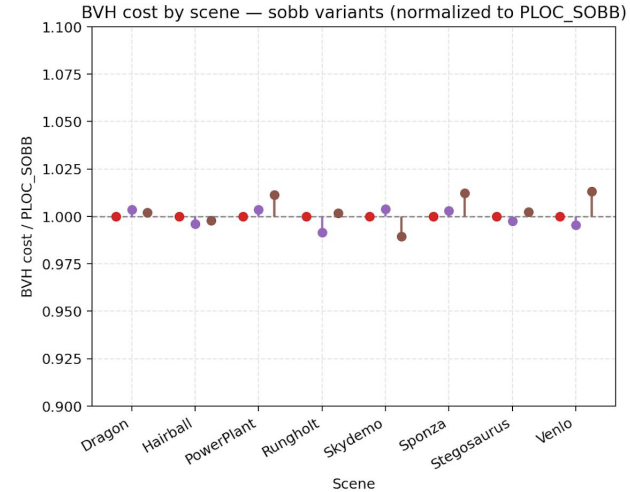
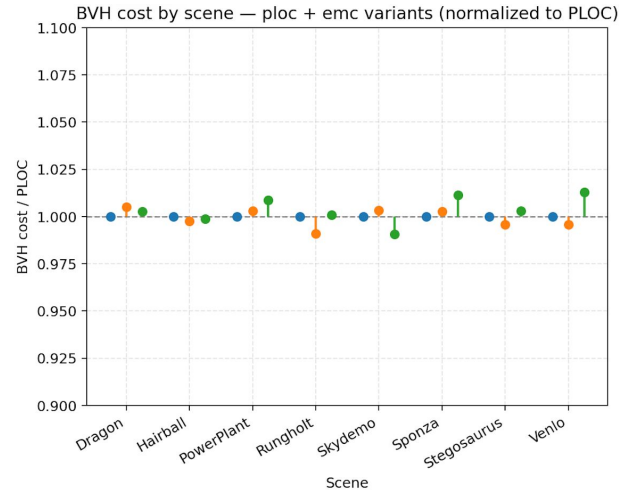
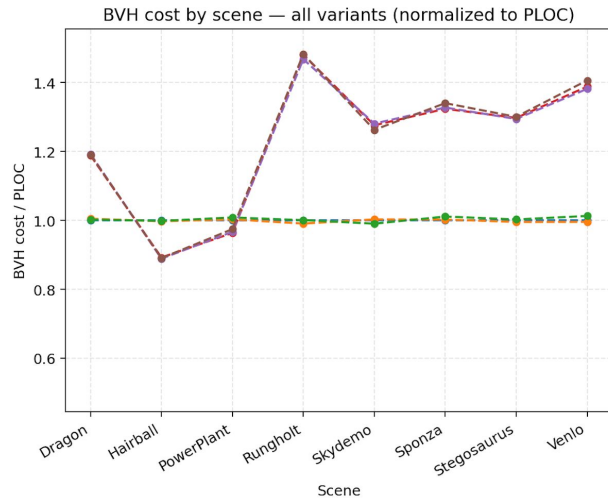
Trace performance by scene — ploc + emc variants (normalized to PLOC)



Trace performance by scene — sobb variants (normalized to PLOC_SOBB)



Results - BVH Cost



Results - Trace time / BVH Cost

Scene	Statistic	PLOC	EMC v1	EMC v2	SOBB	SOBB EMC v1	SOBB EMC v2
Dragon	Time / cost	1.00	1.00	1.09	1.04	1.04	1.14
	Trace [Mrays/s]	1.00	0.99	0.92	0.81	0.81	0.74
Hairball	Time / cost	1.00	1.00	1.01	0.84	0.85	0.85
	Trace [Mrays/s]	1.00	1.00	1.00	1.31	1.31	1.32
PowerPlant	Time / cost	1.00	0.98	0.99	0.87	0.84	0.85
	Trace [Mrays/s]	1.00	1.01	0.99	0.84	0.88	0.86
Rungholt	Time / cost	1.00	1.02	0.97	0.99	1.01	0.96
	Trace [Mrays/s]	1.00	0.98	1.04	0.67	0.66	0.69
Skydemo	Time / cost	1.00	1.11	0.98	0.89	0.92	0.86
	Trace [Mrays/s]	1.00	0.90	1.03	0.91	0.87	0.94
Sponza	Time / cost	1.00	1.02	1.01	1.06	1.08	1.07
	Trace [Mrays/s]	1.00	0.97	0.99	0.72	0.69	0.70
Stegosaurus	Time / cost	1.00	1.01	1.00	0.98	1.00	0.98
	Trace [Mrays/s]	1.00	0.99	1.00	0.79	0.78	0.78
Venlo	Time / cost	1.00	1.00	0.99	0.91	0.91	0.89
	Trace [Mrays/s]	1.00	1.01	1.01	0.79	0.79	0.79

■ **Table 5.12** Normalized ray tracing performance (higher is better) and ray tracing time per BVH cost (lower is better) unit across scenes. All values are relative to the PLOC baseline. Nvidia RTX 4080.

Conclusion

Extended Morton Codes

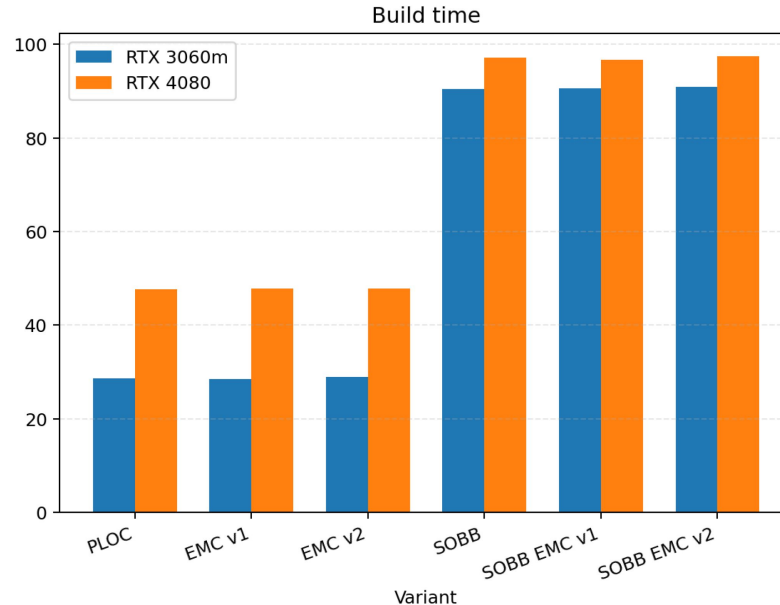
- cheap integration
- minor improvements
- impact on scenes:
 - nonuniform axes lengths
 - nonuniform primitive size

Skewed Oriented Bounding Boxes

- increased build time
- improvement on:
 - complex scenes
 - scenes with nonuniform geometry
- worse performance:
 - all other scenes

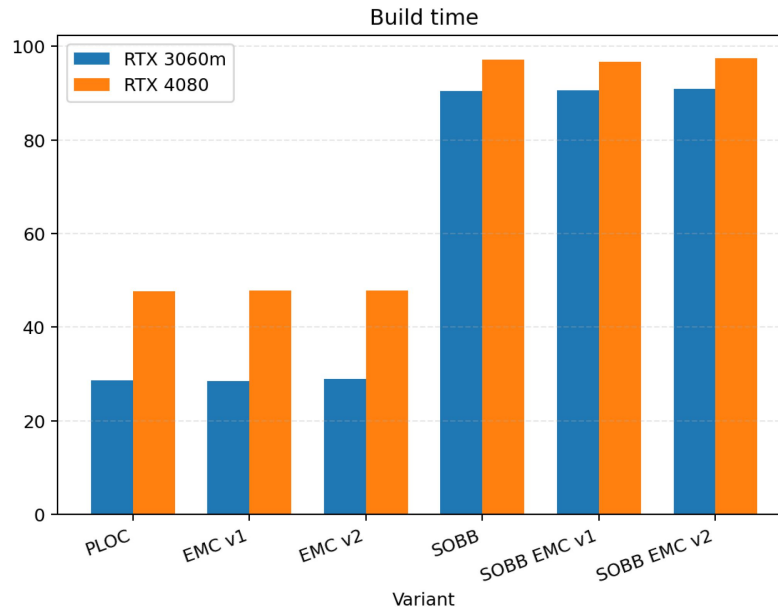
Opponent's question:

BVH build time on the RTX 3060m GPU is faster than on the RTX 4080.
Can you analyze in more detail what is causing this?



Opponent's question:

BVH build time on the RTX 3060m GPU is faster than on the RTX 4080. Can you analyze in more detail what is causing this?



Scene

Dragon

of triangles: 800k

Hairball

of triangles: 2.9M

PowerPlant

of triangles: 12.8M

Rungholt

of triangles: 5.8M

Skydemo

of triangles: 3.7M

Sponza

of triangles: 262k

Stegosaurus

of triangles: 5.7M

Venlo

of triangles: 1.3M

Opponent's question:

BVH build time on the RTX 3060m GPU is faster than on the RTX 4080. Can you analyze in more detail what is causing this?

Scene	PLOC	EMC v1	EMC v2	SOBB	SOBB EMC v1	SOBB EMC v2
Dragon	14.39 (1.00)	14.37 (1.00)	14.34 (1.00)	33.41 (2.32)	33.56 (2.33)	33.63 (2.34)
Hairball	32.24 (1.00)	31.55 (0.98)	32.22 (1.00)	93.70 (2.91)	94.14 (2.92)	94.04 (2.92)
Rungholt	43.10 (1.00)	42.62 (0.99)	43.49 (1.01)	154.24 (3.58)	154.66 (3.59)	155.88 (3.62)
Skydemo	35.67 (1.00)	36.22 (1.02)	36.34 (1.02)	116.29 (3.26)	115.98 (3.25)	116.31 (3.26)
Sponza	9.35 (1.00)	9.57 (1.02)	9.71 (1.04)	16.48 (1.76)	16.69 (1.79)	16.80 (1.80)
Stegosaurus	49.96 (1.00)	49.72 (1.00)	50.27 (1.01)	176.79 (3.54)	177.48 (3.55)	177.30 (3.55)
Venlo	16.41 (1.00)	16.22 (0.99)	16.34 (1.00)	42.25 (2.58)	42.26 (2.58)	42.27 (2.58)

■ **Table 5.3** Build times across scenes with relative values to plain PLOC variant in braces (lower is better). Nvidia RTX 3060 Mobile.

Scene	PLOC	EMC v1	EMC v2	SOBB	SOBB EMC v1	SOBB EMC v2
Dragon	18.86 (1.00)	18.96 (1.01)	18.72 (0.99)	31.03 (1.65)	30.46 (1.61)	30.50 (1.62)
Hairball	44.19 (1.00)	43.10 (0.98)	43.89 (0.99)	78.88 (1.78)	78.86 (1.78)	78.62 (1.78)
PowerPlant	121.62 (1.00)	122.25 (1.01)	121.77 (1.00)	270.96 (2.23)	271.01 (2.23)	271.84 (2.24)
Rungholt	55.97 (1.00)	56.52 (1.01)	56.63 (1.01)	122.23 (2.18)	121.88 (2.18)	121.80 (2.18)
Skydemo	45.30 (1.00)	46.15 (1.02)	46.66 (1.03)	90.82 (2.00)	89.90 (1.98)	90.35 (1.99)
Sponza	15.12 (1.00)	15.24 (1.01)	15.15 (1.00)	20.40 (1.35)	19.83 (1.31)	20.49 (1.35)
Stegosaurus	59.05 (1.00)	59.28 (1.00)	59.42 (1.01)	126.42 (2.14)	126.18 (2.14)	125.62 (2.13)
Venlo	20.22 (1.00)	20.20 (1.00)	20.07 (0.99)	36.51 (1.81)	35.50 (1.76)	36.94 (1.83)

■ **Table 5.4** Build times across scenes with relative values to plain PLOC variant in braces (lower is better). Nvidia RTX 4080.